Theoretical Values (Using Multisim)

Table 3.1T: Data for I - V characteristics.

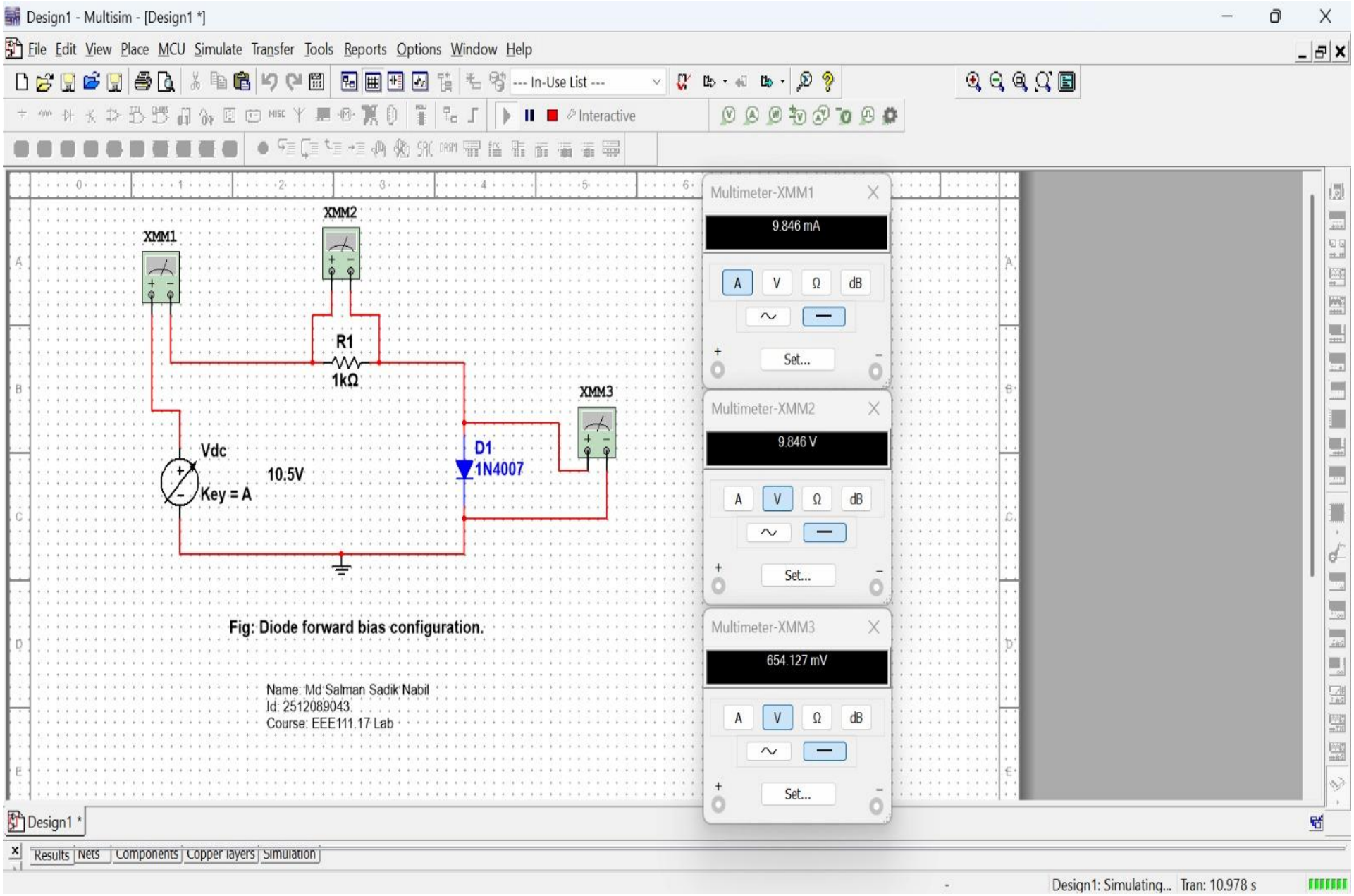
V (volts)	V_R (volts)	V_Z (volts)	Iz = VR / R (ma)
0.1 V	0 V	0.099 V	0 mA
0.3 V	0 V	0.299 V	0 mA
0.5 V	0 V	0.499 V	0 mA
0.7 V	0 V	0.699 V	0 mA
1.0 V	0 V	0.999 V	0 mA
2.0 V	0 V	2 V	0 mA
3.0 V	0 V	3 V	0 mA
4.0 V	0 V	4 V	0 mA
5.0 V	0 V	5 V	0 mA
6.0 V	0.093 V	5.907 V	0.1968 mA
7.0 V	1.044 V	5.956 V	2.221 mA
8.0 V	2.031 V	5.969 V	4.321 mA
9.0 V	3.023 V	5.977 V	6.431 mA
10.0 V	4.017 V	5.983 V	8.547 mA

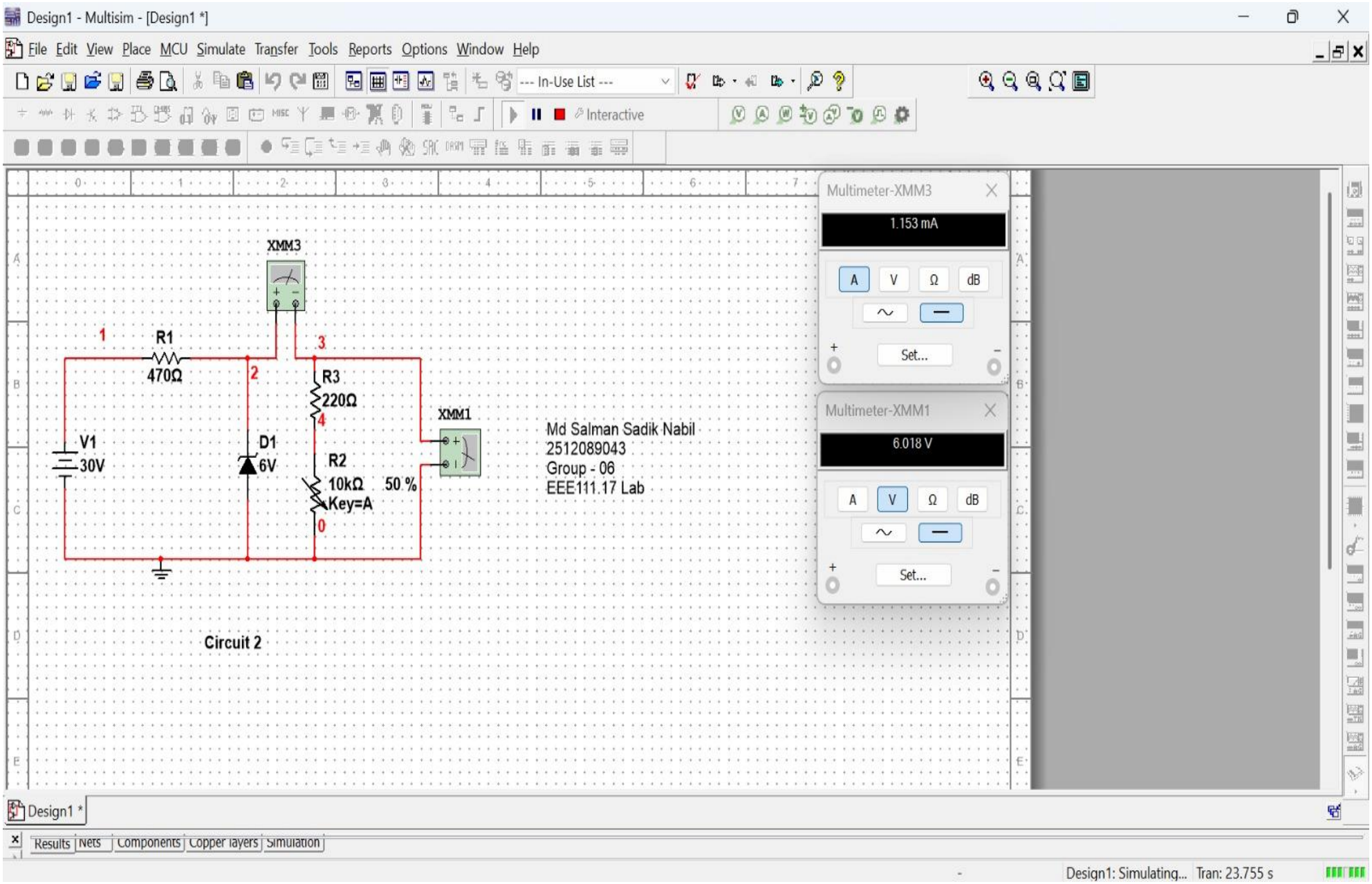
Table 3.2T: Data for Load Regulation

POT R (kΩ)	V220 (mV)	VL (volts)	IL (Amp)= V220/220
1.03 kΩ	986 mV	5.967 V	4.77 mA
2.01 kΩ	589 mV	5.976 V	2.68 mA
3.04 kΩ	403 mV	5.97 V	1.83 mA
5.02 kΩ	251 mV	5.98 V	1.11 mA
6.01 kΩ	211 mV	5.981 V	0.95 mA
7.04 kΩ	181 mV	5.981 V	0.82 mA
9.07 kΩ	141 mV	5.981 V	0.64 mA
10.07 kΩ	127 mV	5.982 V	0.58 mA

Table 3.3T: Data for Line Regulation.

V (volts)	V_L (volts)
1.0 V	0.722 V
3.0 V	2.17 V
6.0 V	4.33 V
8.0 V	5.77 V
9.0 V	5.95 V
10.0 V	5.97 V
11.0 V	5.98 V
12.0 V	5.98 V


Design1 - Multisim - [Design1 *]
File Edit View Place MCU Simulate Transfer Tools Reports Options Window Help
--- In-Use List ---
Interactive
XMM1
XMM2
XMM3
R1
1kΩ
Vdc
10.5V
Key = A
D1
1N4007
Multimeter-XMM1
9.846 mA
A V Ω dB
Set...
Multimeter-XMM2
9.846 V
A V Ω dB
Set...
Multimeter-XMM3
654.127 mV
A V Ω dB
Set...
Fig: Diode forward bias configuration.
Name: Md Salman Sadik Nabil
Id: 2512089043.
Course: EEE111.17 Lab
Design1 *
Results Nets Components Copper layers Simulation
Design1: Simulating... Tran: 10.978 s


Design1 - Multisim - [Design1 *]
File Edit View Place MCU Simulate Transfer Tools Reports Options Window Help
--- In-Use List ---
Interactive
XMM3
R1
470Ω
R3
220Ω
XMM1
V1
30V
D1
6V
R2
10kΩ
50 %
Key=A
Md Salman Sadik Nabil
2512089043
Group - 06
EEE111.17 Lab
Circuit 2
Multimeter-XMM3
1.153 mA
A V Ω dB
Set...
Multimeter-XMM1
6.018 V
A V Ω dB
Set...
Design1 *
Results Nets Components Copper layers Simulation
Design1: Simulating... Tran: 23.755 s